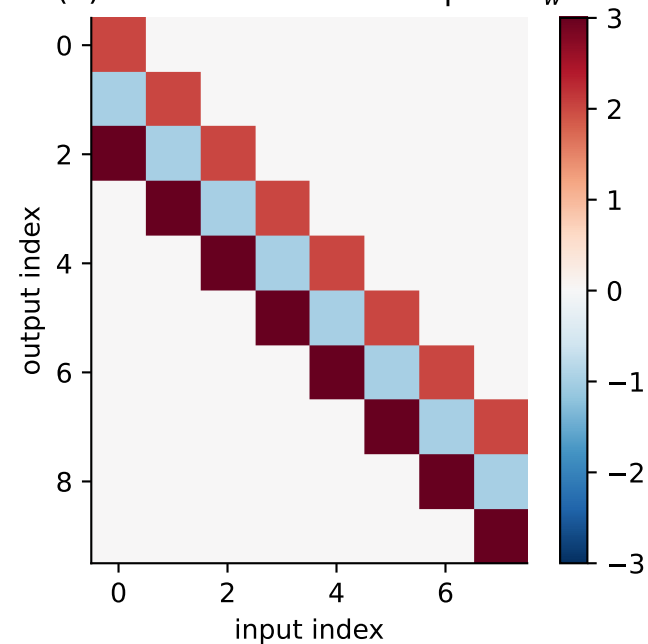
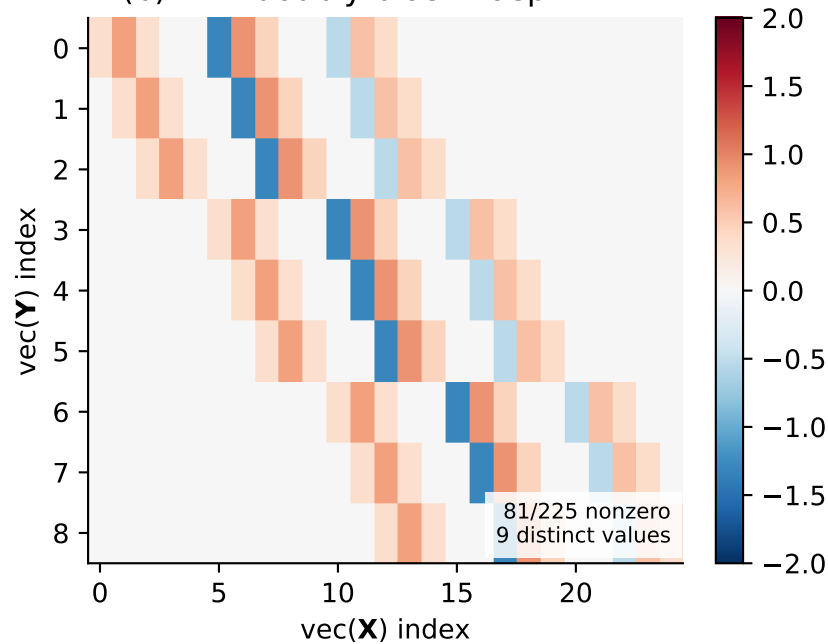


(a) 1-D convolution: Toeplitz $\mathbf{T}_w$



(b) 2-D: doubly block Toeplitz $\mathbf{W}'$



(c) why dense layers do not scale

